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The impact of global warming on ENSO from the perspective of objective signals

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ABSTRACT

Most studies on the relationship between global warming and El Niño-Southern Oscillation (ENSO) rely on simulation experiments, but they have produced contradictory results, which cannot guarantee the reliability of the experimental findings. To address this challenge, we approached the issue from the perspective of objective ENSO and global warming signals. The effects of global warming on ENSO are not limited to the Earth's surface, but also present in the atmosphere. While objective monitoring indices exist for surface global warming and ENSO signals, objective ENSO and global warming signals in the atmosphere are difficult to obtain. To obtain high-precision ENSO and global warming signals in the atmosphere, we propose using the strategy of ICA (Independent Component Analysis) combined with a non-parametric method. The results of the objective ENSO and global warming signals suggest that the global warming signal leads the ENSO signal by approximately three months and exhibits a clear positive correlation with the ENSO signal, especially the occurrence of strong ENSO events is closely related to global warming, indicating that global warming promotes the occurrence of ENSO events. Under the influence of global warming, the frequency and intensity of ENSO increase, the characteristics and complexity of ENSO atmospheric teleconnections may be altered, and there is a greater tendency for more destructive El Niño events to occur. Since the beginning of the 21st century, the probability of ENSO event has increased significantly. Furthermore, moderate or strong ENSO events tend to develop earlier, and extreme ENSO events not only develop earlier but also prolong their decay phase, leading to an extended lifespan of ENSO events. This paper investigates the effect of global warming on ENSO from the perspectives of objective ENSO and global warming signals, which well supplements the objective factual basis for the previous simulation experimental studies.

1. Introduction

El Niño-Southern Oscillation (ENSO) phenomenon and global warming are two significant factors that influence the Earth's climate system. Among them, ENSO originates in the tropical Pacific and is the strongest internal climate mode on interannual timescales (Trenberth, 1997). It not only affects climate anomalies in the tropical Pacific region but also exerts important influences on climate anomalies at tropical and even global scales through atmospheric teleconnections (Cane, 1983; Qadimi et al., 2021; Alizadeh, 2023). ENSO events can be characterized in terms of amplitude, duration, frequency, diversity, and type (Alizadeh, 2017, 2022a, 2022b). Studying the changes of ENSO characteristics is crucial for understanding and predicting climate change. Alizadeh (2022a) analyzed the changes of ENSO amplitude, duration, variability, and frequency during 1959–2021 using ERA5 data, and the results showed that the variability of the amplitude of El Niño events is much higher compared to La Niña events, whereas the variability of El Niño event duration is much lower than La Niña events. In addition, the highest frequency of El Niño events occurs in the fall and winter in the north, while the frequency of La Niña events is highest in the fall in the north. Different types of ENSO events (Eastern Pacific (EP) and Central Pacific (CP) El Niño events) also result in atmospheric teleconnections

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exhibiting different characteristics on a global scale (Alizadeh, 2017). Global warming refers to the long-term increase in the average Earth surface temperature caused by human activities releasing greenhouse gases into the atmosphere (Berger and Tricot, 1992). It is one of the most urgent challenges faced by humanity and the planet. It has profound impacts on ecosystems, economies, and social security, among other aspects. Global warming has already led to glacier melting, sea-level rise, and an increased frequency and severity of extreme weather events (Houghton, 2005).

Understanding the changes of ENSO characteristics under impact of global warming is crucial for comprehending and predicting future climate change. Numerous researchers have conducted studies aided by climate models. Many simulation experiments have indicated that the frequency and intensity of ENSO events undergo changes in the context of global warming (Solomon, and Change, I.P. on C., I, I.P. on C.C.W.G, 2007). The research of Alizadeh indicated that the frequency, amplitude, duration, and diversity of ENSO will change significantly in the future under the influence of global warming based on the latest Coupled General Circulation Models (CGCMs) (Alizadeh, 2022b). Analysis of the Extended Reconstructed Sea Surface Temperature version 5 (ERSSTv5) dataset during the period 1951–2020 indicates that the ratio of the number of CP El Niño to EP El Niño has substantially increased over the past two decades (Alizadeh et al., 2022). Zebiak and Cane (1987), Jin (1997) and Wang and An (2002) have pointed out through modifications of the Zebiak-Cane (ZC) simple coupled model parameters that an increase in the equatorward-averaged thermocline depth in the ZC model leads to shorter ENSO periods and higher frequencies. Collins (2000) using the second-generation Hadley Centre couple model, found that when CO2 concentrations increase fourfold, the meridional thermocline temperature gradient in the equatorial Pacific intensifies, resulting in an accelerated exchange of anomalous heat capacities between the equatorial interior and the surrounding regions, ultimately increasing the frequency of ENSO events. Additionally, Cai et al. (2014a) demonstrated in their research that warming occurred in the equatorial central-eastern Pacific after global warming, leading to an increased occurrence of extreme ENSO events. However, Fedorov and Philander (2001) as well as Wittenberg (2002) hold contrasting viewpoints. Their studies indicate that, in observations and coupled models, when the tropical background zonal winds weaken, the meridional gradient of the thermocline depth decreases, leading to a delayed adjustment of the equatorial thermocline meridional structure. Consequently, after global warming, ENSO periods become longer, and frequencies decrease. Marjani et al. (2019) investigated the changes in the frequency of extreme ENSO under global warming using 14 General Circulation Models (GCMs), and their results suggest that the frequency of extreme ENSO events does not undergo any significant change under future global warming. Similarly, the simulation experiments of Tang et al. (2021) suggests that extreme El Niño frequencies will remain essentially unchanged under the Coupled Model Intercomparison Project phase 5 (CMIP5) warming scenario. Furthermore, regarding ENSO intensity, researchers have obtained opposing conclusions using numerical models. Experimental results from Timmermann et al. (1999), Collins (2000) and Watanabe et al. (2012) suggest that increased CO2 concentrations cause an increase in the equatorial vertical temperature gradient, enhancing the sensitivity of sea surface temperature anomalies to surface wind stress and ultimately resulting in amplified ENSO amplitudes. However, Knutson et al. (1997) argue that increased CO2 concentrations reduce the zonal temperature gradient and thermocline slope of the equatorial climate averages, weaken the ocean surface currents, diminish the vertical oceanic heat flux in the eastern Pacific, and ultimately lead to a weakening of ENSO amplitudes. Additionally, the simulated CO2 increase experiments conducted by Merryfield (2006) revealed that, under a doubling of CO2 levels, three models exhibited an increasing trend in the amplitude of ENSO, whereas five models showed a significant decreasing trend. Considering the contradictory results obtained from a climate model perspective and taking into account the inherent uncertainties and cumulative errors of simulation experiments (Eyring et al., 2016), the authenticity of the findings cannot be guaranteed. Therefore, this study attempts to investigate the mechanisms of the impact between ENSO and global warming from an objective perspective based on ENSO and global warming signals.

The influence of global warming on ENSO extends beyond the surface and is also presented in the atmosphere. Bayr and Latif (2023) studied how the amplification of zonal wind feedback and the damping of heat flux feedback, two atmospheric feedback processes, vary in the dynamics of ENSO under global warming conditions. The results indicate that global warming not only strengthens these two feedback processes, leading to enhanced convection, but also further increases the amplitude of ENSO. The research conducted by Ham (2017) based on the Coupled Model Intercomparison Project Phase 5 models also indicates that the climate induced rise in sea surface temperatures (SST) caused by global warming has an impact on the atmospheric response in the central Pacific during El Niño events. Consequently, studying the relationship between global warming and ENSO in the atmosphere is equally important.

In order to objectively study the relationship between global warming and ENSO, it is necessary to obtain global warming and ENSO signals at both the surface and in the atmosphere. Objective monitoring indices, such as the Global Mean Surface Temperature (GMST) and the Oceanic Niño Index (ONI), can be used to express the surface global warming and ENSO signals. However, there are no objective monitoring indices available to accurately represent the ENSO and global warming signals in the atmosphere. Chen et al. (2018) proposed a method that utilizes Empirical Orthogonal Function (EOF) analysis combined with optimal filtering to extract a reliable ENSORS index from the upper tropospheric specific humidity to describe the ENSO phenomenon. This method provides a feasible approach for this study. To obtain high-precision global warming and ENSO signals, an optimized method is proposed based on Chen et al. (2018), which combines Independent Component Analysis (ICA) with non-parametric. This method effectively overcomes the influence of nonlinear features in climate time series during the extraction of components Mudelsee (2009) and solves the problem of separating information and noise in high-dimensional data.

It is important to determine from which atmospheric component the global warming and ENSO signals should be extracted. Previous studies have shown that water vapor can cause disruptions and provide timely feedback to the climate system in response to global warming and ENSO (Lindzen, 1990; Rind et al., 1991; Pierrehumbert, 1995; Dickinson et al., 1996; Trenberth et al., 2005; Lu et al., 2020). Trenberth et al. (2005) indicated a close relationship between atmospheric water vapor content over the oceans and ENSO events. (Lu et al., 2020) found that the first and second EOF modes of the tropical Convective Precipitation Temperature (CPT) anomalies are associated with the activity of the El Niño-Southern Oscillation (ENSO). On the one hand, Lindzen (1990) and Pierrehumbert (1995) discussed the possible mechanism of decreased upper tropospheric humidity due to global warming. On the other hand, Rind et al. (1991) proposed an argument that global warming leads to increased humidity in the upper troposphere. Dickinson et al. (1996) provided a clear answer that water vapor feedback is positive under global warming. Specific humidity is one of the most commonly used indicators to describe water vapor variations, and it can effectively captures the response of water vapor to ENSO and global warming. Hence, the ENSO and global warming signals can be extracted from specific humidity.

It is equally important to determine the altitude from which the global warming and ENSO signals should be extracted. Considering that this study focuses on using specific humidity to extract the global warming and ENSO signals, the strength of the water vapor feedback will affect the selection of altitude. Existing research has shown that the upper troposphere has a significant impact on surface climate and surface feedback (Gettelman et al., 2011). Changes in upper tropospheric water vapor are crucial for understanding and predicting climate

(Rieckh et al., 2018). Based on the NCEP reanalysis data, Paltridge et al. (2009) studied the trend of specific humidity in the upper troposphere and found that water vapor feedback plays a key role at this altitude. Interestingly, small changes in upper atmospheric specific humidity have a much greater impact on the greenhouse effect compared to the lower atmosphere (Hansen et al., 1984). Vergados et al. (2016) pointed out that the water vapor feedback is strongest at around 250 hPa in the upper troposphere. Tian et al. (2019) indicated that ENSO has a significant impact on the interannual variations of tropical water vapor at upper tropospheric levels. Chen et al. (2018) found that the maximum response to ENSO is observed at the altitude of 250 hPa based on the Constellation Observing System for Meteorology, Ionosphere, and Climate (COSMIC) radio occultation (RO) specific humidity dataset. Previous studies have shown that 250 hPa in the upper troposphere is not only the altitude with the strongest water vapor feedback but also the altitude with the greatest response to ENSO and a larger response to greenhouse effects compared to the lower atmosphere. Consequently, in this study, the objective global warming and ENSO signals will be extracted from the altitude of 250 hPa in the upper troposphere. In summary, this study aims to extract the atmospheric global warming and ENSO signals based on the specific humidity data at the altitude of 250 hPa in the upper troposphere, using the strategy of ICA combined with a non-parametric method. The relationship between global warming and ENSO will be investigated accordingly.

The rest of this paper is organized as follows. In Section 2, ERA 5 specific humidity dataset, indicators expressing objective global warming and ENSO signals and the signal extraction method applied in this paper are interpreted. In Section 3, the objective global warming and ENSO signals in the atmosphere are extracted by the strategy of ICA combined with a non-parametric method, and the relationship between global warming and ENSO in the atmosphere is then analyzed in terms of both the temporal and spatial patterns of these objective signals. In addition, the changes of ENSO frequency, intensity and duration in the context of global warming, and the quantitative relationship between global warming and ENSO were investigated from the perspective of objective ENSO signals. Finally, the corresponding conclusions are presented in Section 4.

2. Datasets and methods

2.1. Datasets

The specific humidity data used in this article stems from the fifth generation of the European Centre for Medium Range Weather Forecasts (ECMWF) atmospheric reanalysis data, which is available in the Earth observation program ECMWF Reanalysis v5 (ERA5) Copernicus Climate Data Store of European Union (https://cds.climate.copernicus.eu). ERA5 dataset is the latest generation of atmospheric reanalysis data generated by using Copernicus Climate Change Service Information following ERA-Interim (Hersbach et al., 2020), which combined with a modern numerical weather prediction model and Observations from a wide range of platforms (insitu, radiosondes, satellite) (Bandhauer et al., 2022). ERA5 dataset is considered the most modern and reliable dataset widely used in long-term analysis and global climate research (Pelosi et al., 2020; Liu et al., 2021; Zhang et al., 2021). ERA5 specific humidity (in g/kg) data is crucial for studying the global water cycle, climate change, and extreme weather events (Hersbach et al., 2020). The high spatial resolution ($0.25^\circ \times 0.25^\circ$) and high temporal resolution (hourly) of ERA5 data make it possible to conduct precise studies of trends and variations in atmospheric water vapor. A long-term time series of ERA5 specific humidity data spanning from January 1959 to January 2021 is selected in this paper to obtain the objective global warming signal in the atmosphere.

2.2. Methods

To ensure the authenticity and reliability of the results, this study further investigates the relationship between global warming and ENSO from the perspective of objective signals, considering the effects of global warming on ENSO reflected in both the surface and the atmosphere. Among them, the global warming and ENSO signals in surface can be expressed through monitoring indices, while the global warming and ENSO signals in atmosphere need to be extracted by the strategy of ICA combined with a non-parametric method. The monitoring indices used to represent the global warming and ENSO signals in this paper and the detailed reasons for utilizing the ICA and non-parametric method are described below.

ENSO, driven by sea surface temperature variations in the central and eastern equatorial Pacific, is an important source of tropical Pacific climate anomalies. It not only has a significant impact on water vapor, but also is one of the most important factors contributing to global climate change. ONI is the most suitable indicator for characterizing ENSO events (Trenberth and Stepaniak, 2001), and it is also one of the most commonly used indices for defining El Niño and La Niña events. Hence, ONI (https://psl.noaa.gov/data/correlation/oni.data) is chosen from the NOAA Physical Sciences Laboratory (PSL) to represent the objective ENSO signal at the surface in this paper. Similarly, global warming is a long-term phenomenon of the widespread increase in the Earth surface temperature caused by greenhouse gases, which has profound impacts on ecosystems, economies, and societies (Stocker, 2014). The Global Mean Surface Temperature (GMST) is one of the most important indicators for measuring global warming and climate change (Valipour et al., 2021). Thus, GMST (https://climate.nasa.gov/vital-signs/global-temperature/) is selected from NASA Goddard Institute for Space Studies (GISS) to represent the objective global warming signal at the surface.

Mudelsee (2009) pointed out that climate system is a complex and nonlinear model. Climate time series own non-Gaussian distributional characteristic. As a consequence, the influence of nonlinear features must be taken into account during the calculation process. Conventional signal processing techniques typically rely on second order statistics, leading to the inevitable neglect of higher-order information in the signal (Jutten and Herault, 1991; Aires et al., 2000). Yet independent component analysis (ICA) originated from the field of Signal Process adopts higher order statistics, which is capable of separating statistically independent components from nonlinear signals (Jutten and Herault, 1991; Comon, 1994), thereby overcoming the influence of nonlinearity (Jutten and Herault, 1991; Aires et al., 2000). In light of this, ICA is used to extract the nonlinear signal of ENSO and global warming in this paper. In addition, Koch and Naito (2007) pointed out that the quality of independent components separated by ICA is affected by the output dimension. If the output dimension is too small, ICA may not find the maximum non-Gaussian direction, leading to information loss. Conversely, if the output dimension is too large, noise will be introduced. Hence, the choice of output dimension has a significant impact on the quality of separated components, and additional post-processing is needed to determine the optimal dimension to retain sufficient information (Lu and Rajapakse, 2006). Currently, the traditional dimension selection method is to retain the number of dimensions that contribute to at least 95% of the total variance. Obviously, traditional method lacks objectivity, and previous studies have also shown that variance contribution is not the determining factor for the optimal dimension (Koch and Naito, 2007). In fact, factors such as outliers and sample size have a greater impact on the selection of dimension. For this reason, this paper adopts a non-parametric or model-free objective method to determine the optimal dimension, using the objective Akaike information criterion (AIC) and data-driven standards to select the optimal dimension, and solves the problem of separating the information and noise of data in high-dimensional problems. In Appendix A, the fundamental principles of Independent Component Analysis (ICA) and non-parametric are given. Similarly, Appendix B presents a detailed flowchart of the strategy of ICA combined with a non-parametric method, along with the comprehensive results of signal changes after each processing step.

3. Results and discussion

3.1. Relationship between ENSO and global warming in the atmosphere

The global warming and ENSO signals in the atmosphere were extracted from the upper tropospheric specific humidity data according to the strategy of ICA combined with a non-parametric method proposed in this paper. The results indicate that, with a lag of 11 months, the extracted IC1 (Independent Component 1) shows a correlation coefficient of 0.910 with GMST. Similarly, with a lag of 8 months, the correlation between IC2 (Independent Component 2) and ONI reaches 0.907. The above findings demonstrate that although there is energy loss in the propagation of the global warming and ENSO signals to the atmosphere (Fedorov, 2007), the signals extracted using the strategy of ICA combined with a non-parametric method are still highly consistent with the original global warming signal (GMST) and ENSO signal (ONI), which confirms the effectiveness of the proposed method in extracting high-quality global warming and ENSO signals in the atmosphere.

Both the temporal and spatial patterns of the extracted global warming and ENSO signals are investigated in this paper to deeply understand the relationship between ENSO and global warming in the atmosphere. To facilitate a comprehensive analysis of the research findings, the years of strong El Niño and La Niña events (The intensity of the ENSO events covered in this paper is given by website https://www.ggweather.com/enso/oni.htm) are highlighted in the Fig. 1 using light red and light blue backgrounds, respectively. It is evident from the temporal sequence of the global warming signal (Fig. 1a) that IC1 or GMST also reached local extremes during years of strong ENSO events, indicating that the occurrence of strong ENSO events is closely related to global warming.

Furthermore, to further elucidate the spatial patterns of the global warming signal and ENSO signal, we have plotted the spatial patterns of IC1 and IC2. The spatial patterns of the global warming signal, as depicted by IC1, exhibit overlaps with the “cold tongue (CT)” structure in the ENSO spatial pattern, which may be attributed to global warming causing an increase in sea surface temperatures, further altering the characteristics of ENSO events (Sun, 2000), thereby enhancing spatial consistency in their response. Moreover, global warming will lead to changes in atmospheric teleconnections associated with ENSO (Yeh et al., 2018). The spatial patterns of the ENSO signal (Fig. 2b) reveal teleconnections between ENSO and the Indian Ocean Dipole (IOD) which coincide with the spatial patterns of the global warming signal (Fig. 2a), exactly suggesting that the atmospheric teleconnection features may be influenced by global warming. Additionally, using a set of different process based indicators, Cai et al. (2014a, 2014b) discovered a significant increase in the frequency of extreme positive IOD events and El Niño events under global warming conditions. Simulation experiments conducted by Endris et al. (2019) also indicate that the current spatial patterns of teleconnections between IOD and ENSO will persist

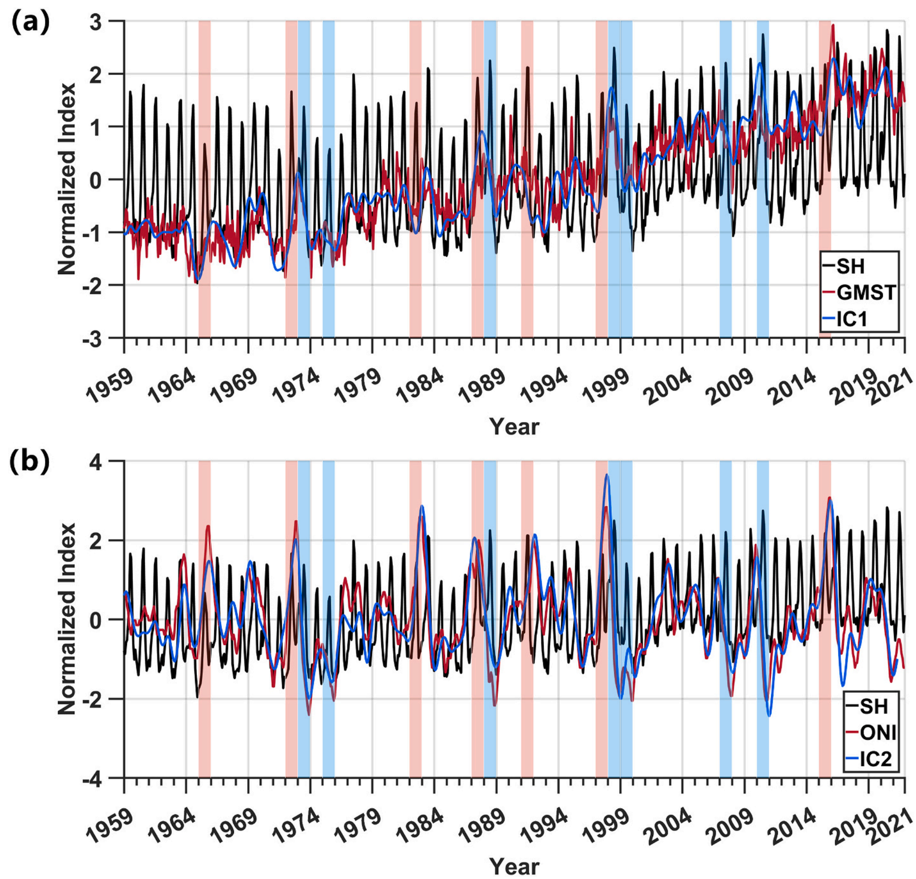

Fig. 1. Signal results in the atmosphere extracted by the strategy of ICA combined with a non-parametric method and the complete signal extraction process is represented in Fig. B2 in Appendix B. (a) Global warming signal (IC1, blue line) extracted by the strategy of ICA combined with a non-parametric method and the time series of GMST (GMST, red line); (b) ENSO signal (IC2, blue) extracted by the strategy of ICA combined with a non-parametric method and the time series of ONI (ONI, red line). The time series of the upper tropospheric specific humidity data are highlighted by the black line. The years of strong El Niño and La Niña events are highlighted using light red and light blue backgrounds, respectively. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

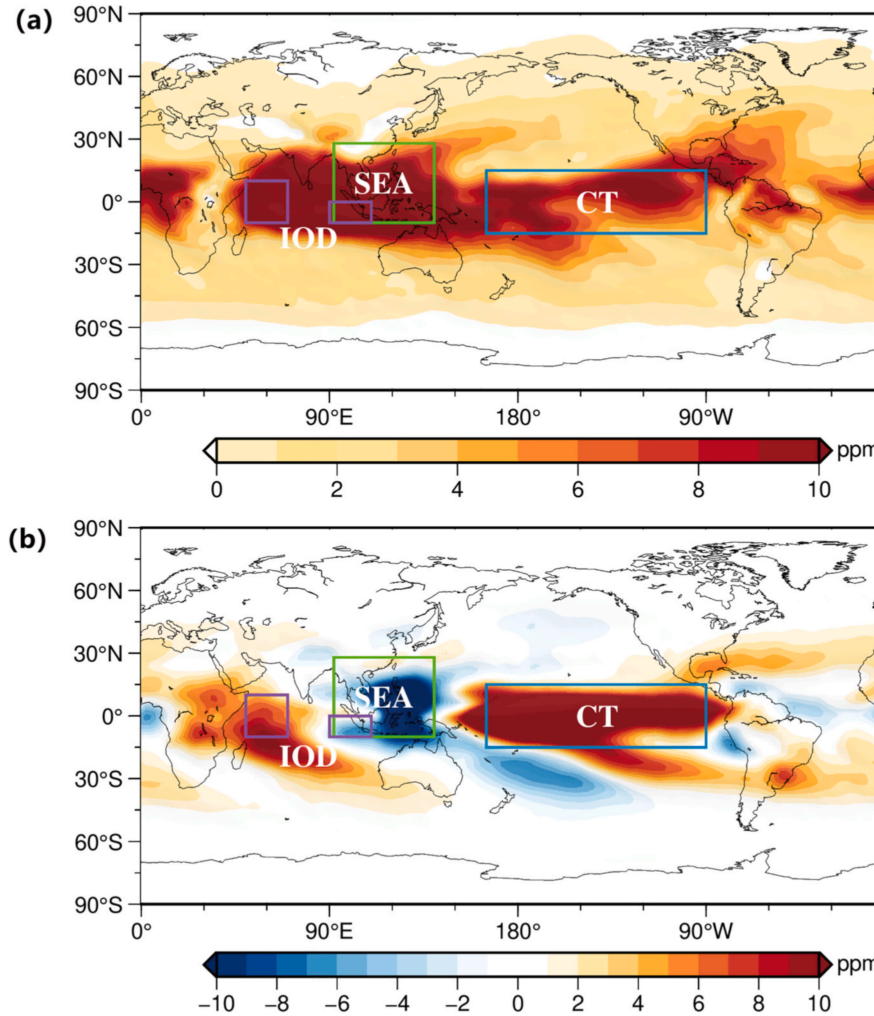

Fig. 2. (a) The spatial pattern of the extracted global warming signal (IC1); (b) The spatial pattern of the extracted ENSO signal (IC2). The area of Indian Ocean dipole (IOD), Southeast Asia (SEA) and cold tongue (CT) are marked with purple, green and blue box respectively. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

under future global warming conditions. Lastly, the spatial patterns of the ENSO signal reveal a cold phase of ENSO in the Southeast Asian (SEA) region, which overlaps with the spatial patterns of the global warming signal. This suggests that global warming may exacerbate extreme climate conditions in Southeast Asia, consistent with the findings of Thirumalai et al. (2017), indicating that global warming significantly influences on the El Niño events in the SEA region and increases the likelihood of extreme events. A possible explanation for the relationship between global warming and ENSO is provided here. Due to the significant role of tropical region in the atmospheric circulation system and its complex climate feedback mechanisms, it exhibits heightened sensitivity to changes in sea surface temperatures. Under the influence of global warming the tropical SST will be affected, which will strengthen the degree of the sea-air coupling phenomenon and leading to anomalous ENSO responses. Moreover, this anomalous ENSO response, influenced by atmospheric circulation patterns, will further alter atmospheric teleconnections. These results demonstrate a clear interaction between ENSO and global warming, indicating that global warming can alter atmospheric teleconnection features, and lead to extreme climate conditions. Moreover, this situation may persist and worsen over time.

3.2. Changes in the characteristics of ENSO under the background of global warming

The previous section of our study has already indicated that global warming may alter the characteristics of ENSO and its atmospheric teleconnections. To gain a clearer understanding of how ENSO features evolve in the context of global warming, we conducted further investigations into the changes in ENSO frequency, intensity, and duration, focusing on objective ENSO signals at the surface.

Global warming is a long-term phenomenon that contributes to a gradual increase in the Earth's surface temperature Naz et al. (2022). Therefore, it is necessary to have a sufficiently long sample length to detect changes in the frequency and intensity of ENSO under a global warming background. However, the record of the surface ENSO signal, as represented by the Oceanic Niño Index (ONI), begins in 1950. The use of ONI alone to study changes in ENSO frequency and intensity is limited by the length of available data. To address this data limitation, we reconstructed the ONI by utilizing SST data (https://psl.noaa.gov/gcos_wgsp/Timeseries/Data/nino34.long.data) from the Niño 3.4 region and applying the same calculation method as the ONI (i.e., three month running mean of SST anomalies in the Niño 3.4 region). The reconstructed index, referred to as Niño $3.4_{\text{Calculated}}$.

The IPCC Fifth Assessment Report (Stocker, 2014) provides relevant descriptions of the temporal evolution of the global average temperature, stating that there was a notable shift in the global average temperature during the 1950s to 1960s, with significant warming observed in the 1960s to 1970s. Considering that the ERA5 specific humidity dataset starts in 1959 and to ensure consistency in the length of the research period before and after global warming and to ensure an adequate sample length, this study adopts 1959 as the dividing line and divides the averaged reconstructed ONI into two periods: the pre-warming period (Pre-GW, 1896.1–1958.12) and the later warming period (Late-GW, 1959.1–2021.12). The correlation between the reconstructed results and the ONI is as high as 0.985, indicating that the reconstructed ONI is reliable and can be used to analyze the effects of global warming on the changes in ENSO characteristics (Fig. 3).

To investigate the changes in the frequency of ENSO occurrences under global warming, this study conducted a statistical analysis based on the reconstructed ONI, including the calculation of ENSO event counts every five years (Fig. 4a) and the distribution histogram of reconstructed ONI values for the Pre-GW and Late-GW (Fig. 4b). In Fig. 4a, years with ENSO event counts equal to the average (about 3 times) are defined as “moderate years” (grey shading), and years above the average are defined as “high-frequency years” (red shading), while years below the average are defined as “low-frequency years” (blue shading).

From the Fig. 4a, it is evident that under the influence of global warming, the probability of high-frequency ENSO years has significantly increased, while the probability of low-frequency ENSO years has substantially decreased. Since the low-frequency years of 1956–1961 in which ENSO events occurred in 1957 and 1958 respectively, this low-frequency years are considered to be in the Pre-GW. The occurrence of low-frequency years decreased from seven occurrences during the Pre-GW to zero occurrences during the Late-GW, while the occurrence of high-frequency years increased from two occurrences during the Pre-GW to four occurrences during the Late-GW. Every year from 1951 to 1956 experienced an ENSO event, which could be associated with the statistically significant warming trend observed globally since the 1950s (Duan et al., 2006). Similarly, from 1971 to 1976, ENSO events occurred every year, which may be linked to the intensified global warming observed in the 1970s (Kosaka and Xie, 2013). The high frequency of ENSO events during the periods of 1996–2001, 2006–2011, and 2016–2021 could be attributed to the sustained global warming trend in the 21st century (Huang et al., 2017). Statistical analysis also indicates that there were a total of 31 ENSO events during the Pre-GW (1896.1–1958.12), whereas there were 43 ENSO events during the Late-GW (1959.1–2021.12), representing a growth rate of 38.71%. Furthermore, since the 21st century, there have been 15 ENSO events, with a high occurrence probability of 75%, significantly higher compared to the occurrence probability of ENSO events (49.2%) during the Pre-GW. The distribution histogram of reconstructed ONI values in Fig. 4b shows that during Pre-GW, the values were mostly within the [−2,2] range (Table 1). However, during the Late-GW, there were even three months (i.e., 2015/10,11,12) with values in the [2.5, 3] range which is related to an extremely intense ENSO event in 2015/16, indicating that since the late stages of global warming, particularly during the 21st century, the intensity of extreme ENSO events has reached its highest level on record, owing to the impact of global warming. Moreover, the analysis reveals that after experiencing global warming, the duration of the ENSO cold phase [−2, −0.5] increased from 144 months to 182 months, while the duration of the ENSO warm phase [0.5, 3] increased from 193 months to 203 months. The increase was particularly prominent for strong El Niño events [1.5, 3] and strong La Niña events [−1.5, 2]. These results demonstrate that under global warming, the frequency of ENSO events has significantly accelerated, and the probability of extreme ENSO events has increased.

Extreme ENSO events can have devastating impacts on global climate and biodiversity (Cai et al., 2014a). As a consequence, it is crucial to study whether greenhouse warming affects the intensity of ENSO events. ENSO exhibits seasonality, typically developing during spring and summer, maturing during autumn and winter, and peaking in winter, followed by a decline in the subsequent spring and summer (Rasmusson and Carpenter, 1982). To investigate the changes in ENSO intensity before and after global warming, the reconstructed ONI during winter was used to characterize ENSO intensity. Fig. 5 presents the intensity of El Niño and La Niña events during the Pre-GW and Late-GW. It can be observed that during the Pre-GW, El Niño event intensity showed a gradual decrease, while La Niña event intensity exhibited a gradual increase. In contrast, during the Late-GW, the situation was reversed. This may be attributed to the shallowing of the thermocline in the Niño3.4 region, which enhances vertical exchange processes (Lee and McPhaden, 2010). Additionally, extreme ENSO events with intensities exceeding 2 °C occurred during the Late-GW, and the intensity of these extreme events has been gradually increasing. These results indicate that under the influence of global warming, there is causing an increase in El Niño events and a relative decrease in La Niña events, suggesting a greater tendency for ENSO events to occur with more damaging El Niño events. Furthermore, the intensity of extreme ENSO events continues to increase.

To further validate the conclusion that ENSO characteristics change under global warming, this study conducted wavelet analysis to examine the frequency domain of the reconstructed ONI, as shown in Fig. 6. The

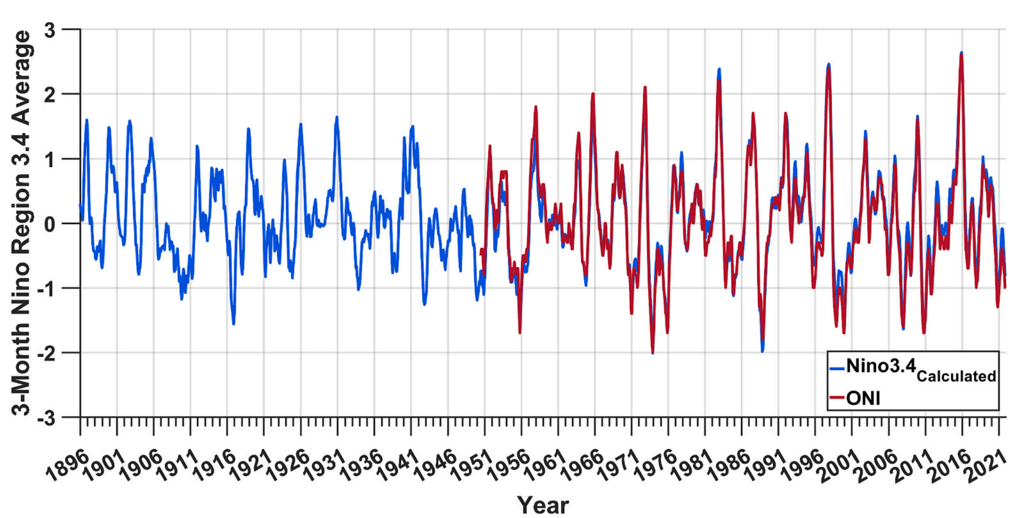

Fig. 3. The reconstructed ONI (blue line) based on the SST of the Niño 3.4 region. The time series of ONI published by official agencies (red line). (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

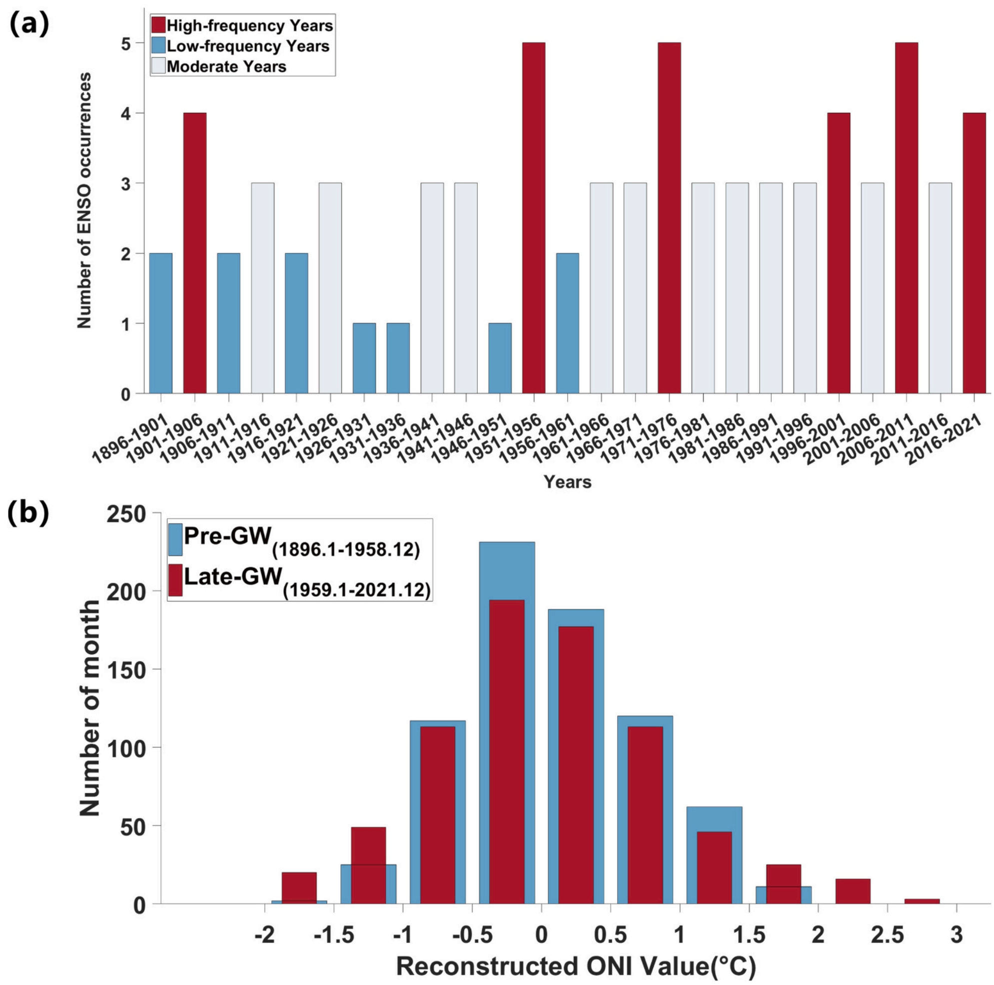

Fig. 4. (a) Statistical histogram of the number of ENSO occurrences per 5 years based on the reconstructed ONI. “High-frequency years”, “low-frequency years” and “moderate years” are indicated by red, blue and white shading, respectively; (b) The distribution histogram of reconstructed ONI values, and all 756 samples in each period are distributed into 0.5 °C bins, for the Pre-GW (blue) and Late-GW (red) periods. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

Table 1
The number of months under different distribution interval of reconstructed ONI values.

Stage	Distribution interval (°C)									
	[−2, −1.5]	[−1.5, −1]	[−1,0.5]	[0.5,0]	[0,0.5]	[0.5,1]	[1,1.5]	[1.5,2]	[2,2.5]	[2.5,3]
Pre-GW	2	25	117	231	188	120	62	11	0	0
Late-GW	20	49	113	194	177	113	46	25	16	3

solid lines in the plot enclose the significant peaks in the power spectrum, determined through a 95% significance test. Moreover, to comprehensively analyze the relationship between wavelet power spectrum and ENSO events, the intensities of historical ENSO events were marked on the corresponding years of the coordinate axis. From the wavelet power spectrum of the reconstructed ONI (Fig. 6a-c), it can be observed that the recorded ENSO events correspond to separate peaks in wavelet power. Additionally, the high-power spectrum region in the plot predominantly falls within the 2–7 year wave band, consistent with the fact that ENSO has a 2–7 year period (Zhang et al., 2022). Furthermore, it was found that years with peaks in wavelet spectra occurring at periods <2 year mostly correspond to strong or moderate El Niño/La Niña years, such as 1957–58 (Strong El Niño), 63–64 (Moderate El Niño), 65–66 (Strong El Niño), 72–73 (Strong El Niño), 73–74 (Strong La Niña), 75–76 (Strong La Niña), 82–83 (Very Strong El Niño), 87–88 (Strong El Niño), 88–89 (Strong La Niña), 97–98 (Strong El Niño), 98–99 (Strong La Niña), 2007–08 (Strong La Niña), 09–10 (Moderate El Niño), 10–11 (Strong La Niña), and 15–16 (Strong El Niño). This may be attributed to the advance occurrence and development of strong or moderate El Niño/La Niña events during the Late-GW, resulting in a shorter interval between consecutive ENSO events and the peaks in wavelet spectra of ENSO signals appearing in the wave band with a period of <2 year. Additionally, it was observed that years in which the entire peaks in wavelet spectra exceeded the 2–7 year wave band range corresponded to the most extreme El Niño/La Niña years in the historical record, such as 73–74 (Strong La Niña), 98–99 (Strong La Niña), and 15–16 (Very Strong El Niño). This may be related to the findings of Sun et al. (2018) that extreme ENSO events occur and develop earlier during the Late-GW, with a prolonged decay phase and significantly extended lifespan (Sun et al., 2018). These results indicate that, overall, most ENSO events fall within the 2–7 year period range. However, against the backdrop of global warming, the occurrence of moderate/strong ENSO

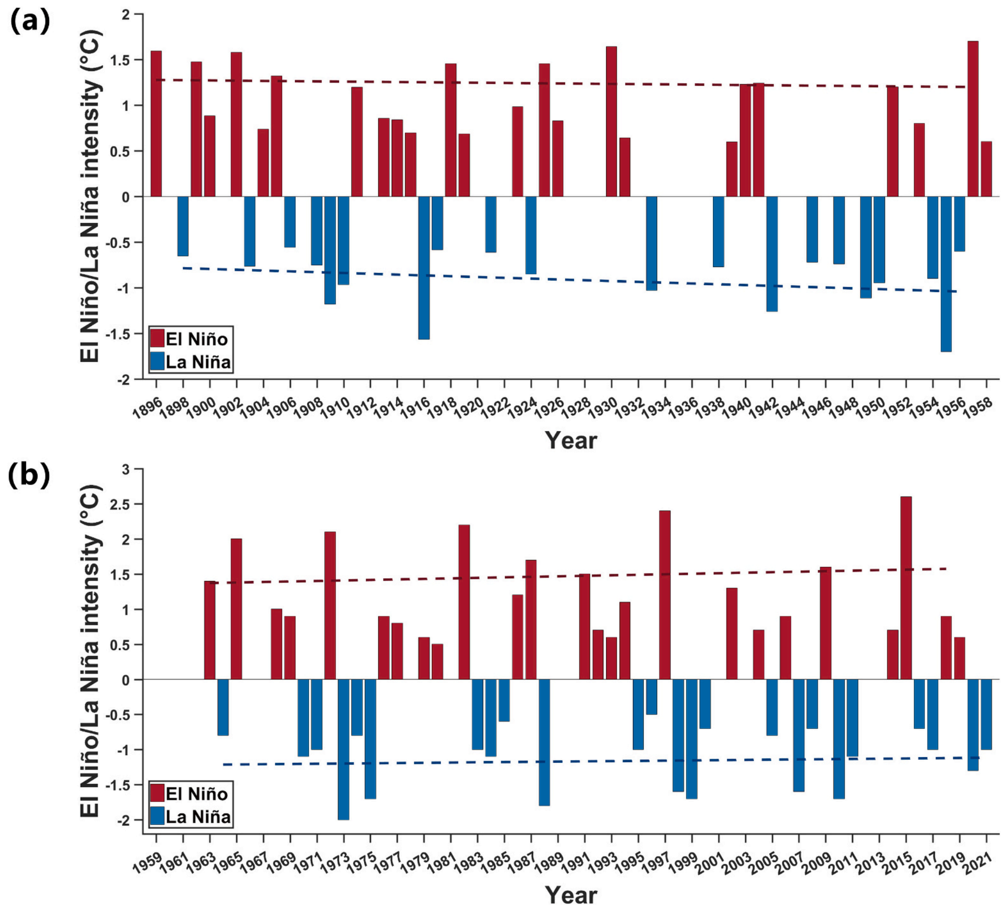

Fig. 5. (a) Intensities of El Niño and La Niña events in the Pre-GW. (b) Intensities of El Niño and La Niña events in the Late-GW. The red dashed line and the blue dashed line represent the linear trend under El Niño and La Niña events, respectively, which is derived from the linear trend calculated from the reconstructed ONI in winter under El Niño or La Niña events. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

events has significantly advanced, and the lifespan of extreme ENSO events has also increased.

The global power spectrum is obtained by averaging the wavelet power spectrum over time (Kestin et al., 1998) and can be used to identify dominant frequencies, periodicity, and trends in the signal. The Fig. 6d presents the global power spectra for the Pre-GW, Late-GW, and the entire time span. Overall, the three global power spectra exhibit similar changing patterns, characterized by three distinct peaks, with the maximum power (dominant frequency) reaching approximately a 4-year period. This is consistent with the findings of Neelin et al. (1998) regarding the spectral frequency of ENSO events peaking at approximately 3–5 years (Neelin et al., 1998). Although the global power spectra of different periods share similar changing patterns, the periods corresponding to the spectral peaks differ. Table 2 provides the periods corresponding to the peak of the local power spectrum in the global power spectrum curve for the three periods Total, Pre-GW and Late-GW. Table 2 reveals that the periods associated with the three spectral peaks are the shortest in the Late-GW, the longest in the Pre-GW, and intermediate for the entire time span. These results indicate that the ENSO period in the Late-GW has significantly shortened compared to the Pre-GW, reflecting a notable increase in the frequency of ENSO events in recent years.

3.3. Quantitative analysis of the relationship between ENSO and global warming signals

Quantitative analysis of the relationship between ENSO and global warming signals is crucial for gaining a deeper understanding of their interaction. In this study, we utilized two widely used methods, correlation analysis and wavelet coherence, to quantitatively analyze the relationship between global warming and ENSO signals.

Global warming is a complex process influenced by multiple factors. Existing evidence suggests that the greenhouse effect, which can be expressed by changes in CO_2 concentration (https://gml.noaa.gov/ccgg/trends/), is the primary driver of global warming. This conclusion is supported by Fig. 7. To investigate the mutual interaction between global warming and ENSO, the global warming signals IC1 and GMST were analyzed after removing the influence of CO_2 concentration, as shown in Fig. 8. The results indicate a correlation of 0.469 between GMST (after CO_2 removal) and ONI, with a lead time of 3 months for GMST relative to the ONI. This suggests that global warming, affecting abnormal sea surface temperatures, plays a role in promoting the occurrence of ENSO phenomena. The improved correlation of 0.629 between IC1 (after CO_2 removal) and ONI suggests a closer influence between the global warming and ENSO signal in the atmosphere. In addition, the correlation between IC1noCO2 (after CO_2 removal) and IC2 is also notably high at 0.501, with a lead time of 6 months for IC1noCO2 relative to IC2, show that despite energy the loss during the propagation process of ENSO signals, there persists a robust correlation and leading response. This also reflects the direct influence between the global warming signal and the ENSO signal in the atmosphere. It is evident from the graph, especially during strong El Niño/La Niña years, that the global warming signal (after CO_2 removal) is significantly correlated with the ENSO signal, aligning with local peaks such as in 1965–1966, 1972–1973, 1973–1974, 1975–1976, 1982–1983, 1987–1988, 1988–1989, 1997–1998, 1998–1999, 1999–2000,

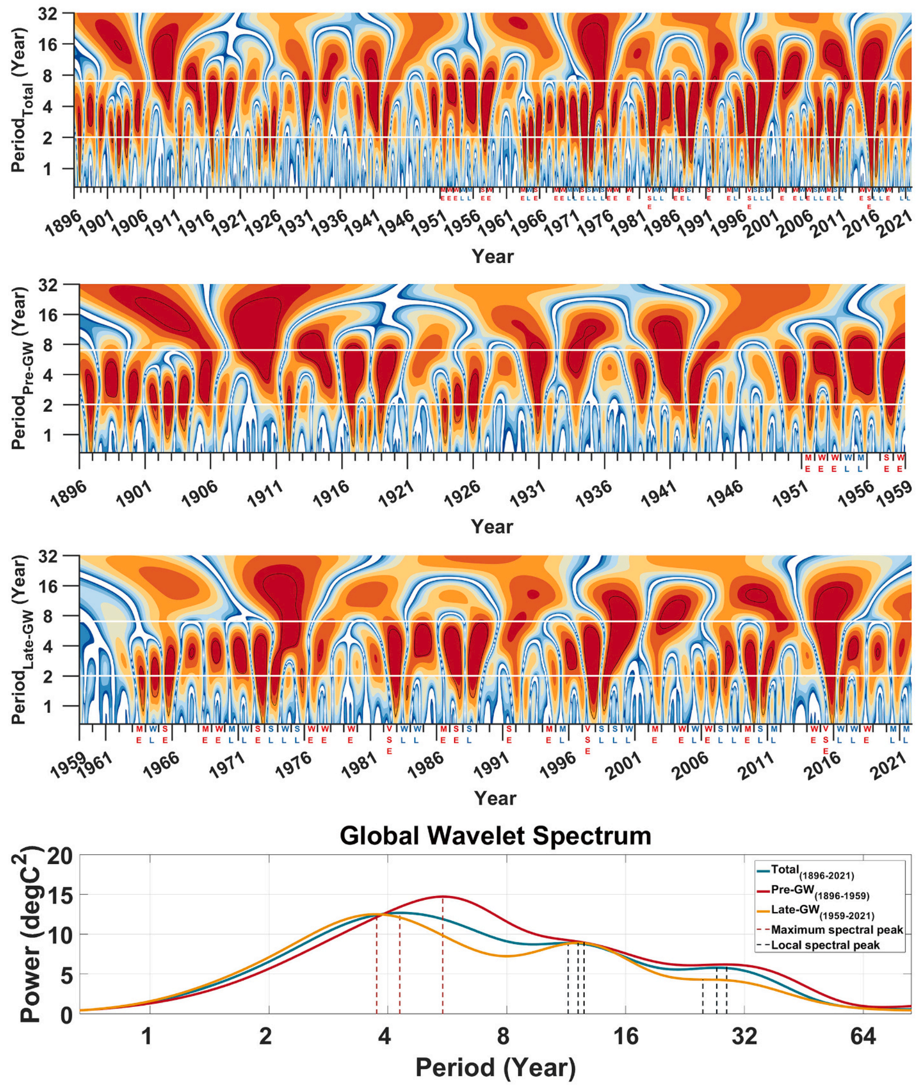

Fig. 6. (a) Wavelet power spectrum of the entire time span; (b) Wavelet power spectrum of the Pre-GW; (c) Wavelet power spectrum of the Late-GW; (d) Global wavelet spectrum of the Pre-GW (red line), Late-GW (orange line), and the entire time span (green line). The red dashed line and the black dashed line represent the periods corresponding to the maximum and local peaks in the global power spectrum, respectively. In (a) (b) (c), the part of the graph surrounded by two white lines is the periodic band of 2–7 year. And the intensities of historical ENSO events were marked on the corresponding years of the coordinate axis of wavelet power spectrum, where WE = Weak El Niño, ME = Moderate El Niño, SE = Strong El Niño, VSE = Very Strong El Niño, WL = Weak La Niña, ML = Moderate La Niña, SL = Strong La Niña. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

Table 2
The period corresponding to the peak of the global wavelet spectrum in Fig. 6d under different stage.

Stage	Period corresponding to the peak of the spectrum		
	Peak 1	Peak 2	Peak 3
Total	4.29 yr	12.12 yr	27.22 yr
Pre-GW	5.51 yr	12.55 yr	28.84 yr
Late-GW	3.75 yr	11.44 yr	25.10 yr

2007–2008, 2010–2011, and 2015–2016. These results indicate a significant positive correlation between the global warming signal and the ENSO signal, and thus global warming enhances the occurrence of ENSO phenomena, especially when the IC1noCO2 (Signal of IC1 after subtracting out the effect of CO2) signal sudden increases, often leads to the occurrence of strong ENSO events.

To quantitatively analyze the relationship between global warming and ENSO in the frequency domain, a wavelet coherence analysis was conducted on the signals of global warming and ENSO, as shown in

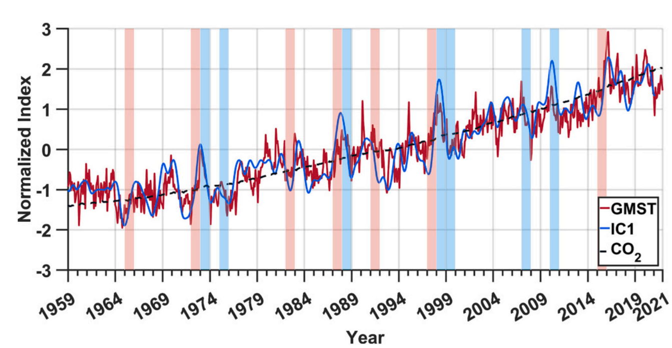

Fig. 7. The time series of objective GMST indicators (red line), the extracted global warming signal (IC1) (blue line) and CO_2 concentration (black dashed line). (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

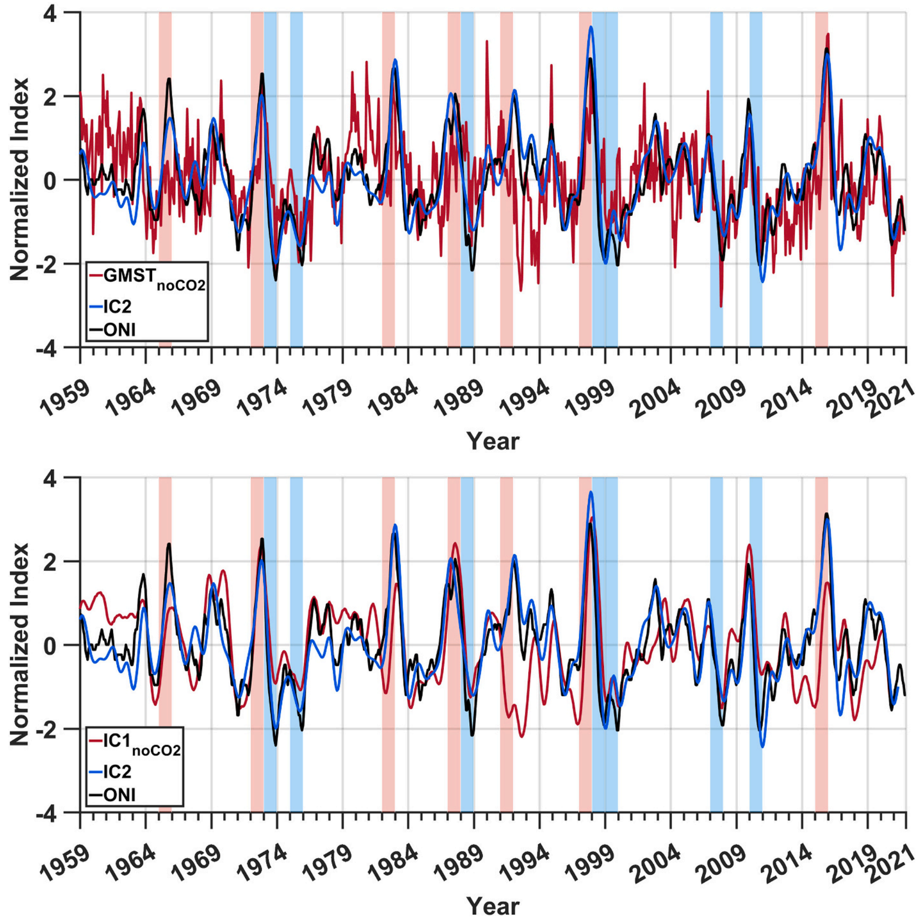

Fig. 8. (a) The time series of objective GMST indicators after removing the effects of CO_2 (red line), the extracted ENSO signal (IC2) (blue line) and ONI published by official agencies (black line); (b) The time series of the extracted global warming signal (IC1) after removing the effects of CO_2 (red line), the extracted ENSO signal (IC2) (blue line) and ONI published by official agencies (black line). (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

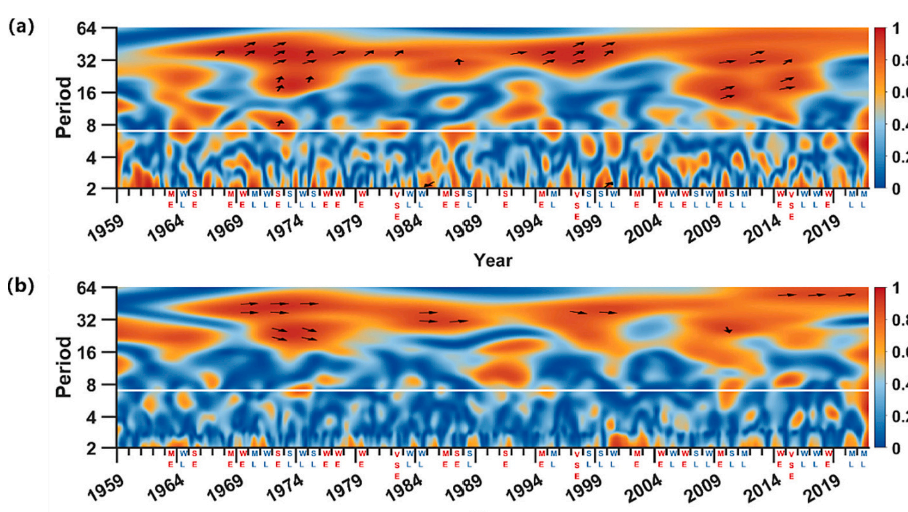

Fig. 9. (a) The wavelet coherence power spectrum between ONI and GMST. (b) The wavelet coherence power spectrum between IC2 and GMST. The part of the graph below the white line is the periodic band of 2–7 year. The intensities of historical ENSO events were marked on the corresponding years of the coordinate axis of wavelet power spectrum, where WE = Weak El Niño, ME = Moderate El Niño, SE = Strong El Niño, VSE = Very Strong El Niño, WL = Weak La Niña, ML = Moderate La Niña, SL = Strong La Niña. Relative phase relationships are shown as arrows (in-phase pointing to the right, inverted phase pointing to the left, an upward pointing arrow indicates that the latter time series is ahead of the former time series, a downward pointing arrow expresses the opposite situation).

Fig. 9. When the coherence wavelet correlation coefficient between two time series is high, the relative lead (lag) of the latter time series with respect to the former can be estimated by analyzing the angle between the direction of the coherence wavelet phase arrow and the time axis. It can be observed that the significant correlations in the wavelet coherence power spectrum mainly exist in the 16–64 year scale, indicating that the mutual influence between global warming and ENSO is primarily manifested at longer timescales. Furthermore, the results demonstrate a positive correlation between ONI (IC2) and GMST, with the majority of arrows in Fig. 9 located in the first quadrant, indicating that GMST leads ONI and confirming the significant positive correlation between the global warming signal and the occurrence of ENSO phenomena, as well as the enhancing effect of global warming on ENSO, as concluded earlier. Moreover, during strong El Niño/La Niña years, GMST and ONI mostly exhibit strong positive correlations (i.e., there is an arrow when the coherent wavelet correlation coefficients of the two time series have a high value), implying that global warming plays a proactive role in the occurrence of extreme ENSO events. In addition, comparing Coherent Wavelet Analysis between IC2 and GMST, as well as ONI and GMST, reveals a stronger correlation within the 2–7 year frequency band for ONI and GMST, which may be attributed to the short-term variations influenced by global warming on sea surface temperature and temperature gradients (Li et al., 2021).

4. Discussion and conclusions

ENSO, as a prominent tropical Pacific signal, exerts significant impacts on global interannual climate anomalies and has profound consequences for extreme weather events and human socioeconomic systems (Cane, 1983). Understanding how ENSO evolves and changes under the background of global warming is a crucial frontier in climate change research. Most studies on the impact of global warming on El Niño-Southern Oscillation (ENSO) rely on simulation experiments (Latif and Keenlyside, 2009; Marjani et al., 2019; Callahan et al., 2021; Alizadeh, 2022b; Lopez et al., 2022), but they have produced contradictory results (Zheng et al., 2016; Chen et al., 2017; Beobide-Arsuaga et al., 2021), which cannot guarantee the reliability of the experimental findings. To provide the objective factual basis for the previous simulation experimental studies, this paper investigates the effect of global warming on ENSO from the perspectives of objective ENSO and global warming signals. Given that the influence of global warming on ENSO involves both the surface and the atmosphere (Ham, 2017; Bayr and Latif, 2023), it is crucial to obtain the objective signals expressing ENSO and global warming in the surface and the atmosphere. While objective monitoring indices exist for surface global warming and ENSO signals (e.g., Global Mean Surface Temperature (GMST), Oceanic Niño Index (ONI)), there is no specific index to express objective ENSO and global warming signals in the atmosphere. Accordingly, we propose the strategy of ICA combined with a non-parametric method to extract the objective signals of global warming and ENSO in the atmosphere. This study reveals the effects of global warming on ENSO characteristics based on objective ENSO and global warming signals in terms of statistics, quantitative analysis, and signal correlation, thus providing an important scientific basis for understanding the changes of ENSO characteristics.

Although there is energy loss during the propagation of ENSO (Fedorov, 2007), the proposed method in this study successfully extracts the high-precision and accurate signals of global warming and ENSO in the atmosphere. With an 11-month lag, the extracted global warming signal (IC1) has a correlation coefficient of 0.910 with GMST. Similarly, with an 8-month lag, the extracted ENSO signal (IC2) has a correlation coefficient of 0.907 with ONI. The previous studies have found ENSO responses to global warming in the "cold tongue" region (Sun, 2000), the IOD region (Cai et al., 2014b), and the SEA region (Thirumalai et al., 2017) of the ENSO signal. Simultaneously, it has been observed that global warming alters the atmospheric teleconnection, thereby further contributing the changes in ENSO characteristics (Yeh et al., 2018). To validate previous findings and provide objective scientific evidence, this study investigates the relationship between global warming and ENSO by examining the temporal and spatial patterns of objective ENSO and global warming signals. Objective signal results indicate that during the years with strong ENSO events, the global warming signal (IC1 or GMST) reaches local extrema from the time series of global warming signals, indicating a close relationship between the occurrence of strong ENSO events and global warming. At the same time, the spatial pattern results demonstrate overlaps between the spatial pattern of the global warming signal and the "cold tongue" structure in ENSO, the Indian Ocean Dipole (IOD), and the ENSO cold phase region in Southeast Asia. The above temporal and spatial results show that global warming may strengthen the degree of the sea-air coupling phenomenon and further lead to alter the characteristics of atmospheric teleconnections, leading to increased complexity in atmospheric teleconnections associated with ENSO under the global warming background.

Although the frequency, amplitude, duration, and diversity of ENSO will change significantly in the future under the influence of global warming based on the latest CGCMs (Alizadeh, 2022b), research on the impact of global warming on the change of ENSO characteristics based on the facts of objective signals is relatively vacant. Global warming is a long-term phenomenon that contributes to a gradual increase in the Earth surface temperature (Naz et al., 2022). Therefore, it is necessary to reconstruct the ONI based on sea surface temperatures in the Niño 3.4 region to ensure a sufficiently long sample length to detect changes in the frequency and intensity of ENSO under a global warming background. The reconstructed ONI is highly consistent with the objective ENSO signal, and the correlation between them is as high as 0.985, so it can be used instead of the objective ENSO signal. The frequency, intensity and duration are three important characteristics used to describe ENSO events (Alizadeh, 2017). This study conducted statistical analyses of the changes in these ENSO characteristics based on objective ENSO signals at the surface (reconstructed ONI index). Cai et al. (2014a) used CMIP5 models and conclude that the number of extreme El Niño events more than doubles under future global warming, while the results of Marjani et al. (2019) and Tang et al. (2021) based on the same generation of CGCMs do not show any considerable change in the frequency of extreme El Niño events. Here, our results are in agreement with the research of Cai et al. (2014a), indicating a significant increase in the occurrence probability of ENSO and extreme ENSO events in the context of global warming. In particular, since the 21st century there have been 15 ENSO events with a high occurrence probability of 75%, significantly higher compared to the occurrence probability of ENSO events (49.2%) during the Pre-GW. This result is consistent with the increase in ENSO frequency since 1999/2000 noted by Hu et al. (2020). Extreme ENSO events can have devastating impacts on global climate and biodiversity (Cai et al., 2014a). It is crucial to study the changes in ENSO event intensity. Cai et al. (2021) argue that ENSO events will strengthen under the global warming forcing in the 21st century, while the results of Callahan et al. (2021) indicate that global warming is associated with the weakening of ENSO. To address disagreement, this study measured the changes in ENSO intensity from the objective signal perspective. Our results show that there is causing an increase in El Niño events and a relative decrease in La Niña events under the influence of global warming. Additionally, the intensity of these extreme events has been gradually increasing, which is consistent with Ham (2018) who noted that the amplitude of El Niño events increases under global warming. This may be attributed to the shallowing of the thermocline in the Niño3.4 region, which enhances vertical exchange processes (Lee and McPhaden, 2010). Alizadeh (2022b) pointed out that the existing models do not simulate the developmental processes of ENSO well and the results regarding the duration of ENSO are not reliable. Here, this study employed wavelet analysis to analyze the changes in the duration of ENSO based on the objective ENSO signal for the first time. The wavelet power spectrum results show that the occurrence of moderate/

strong ENSO events has significantly advanced, and the lifespan of extreme ENSO events has also increased against the backdrop of global warming. This is consistent with Wu et al. (2019) point that the duration of ENSO is generally related to the amplitude of ENSO. On the other hand, the duration of ENSO is also related to the timing of the onset of the event (Lee et al., 2014; Wu et al., 2019, 2021), but this phenomenon did not appear in our results.

Most of the previous studies on the impact of global warming on ENSO have been statistical qualitative analyses based on simulation experiments (Cai et al., 2014a, 2018; Marjani et al., 2019; Tang et al., 2021), and there is still a lack of studies on the quantitative analysis of the impact of global warming on ENSO from the perspective of objective signals. In this study, correlation analysis and wavelet coherence analysis are employed to further quantify the impact of global warming on ENSO. The results of the correlation analysis demonstrated a significant positive correlation between the global warming signal and the occurrence of El Niño events, with the global warming signal leading the ENSO signal. Therefore, global warming plays a promoting role in the occurrence of ENSO events by inducing anomalous sea surface temperatures, particularly leading to intense El Niño events when the global warming signal experiences sudden increases. The above correlation results are consistent with the findings of Sun et al. (2018), indicating that global warming has a contributing effect on the occurrence of ENSO, especially causing a significant increase in the frequency of extreme La Niña events and extreme El Niño events. The findings of wavelet coherence analysis revealed that the interaction between global warming and ENSO primarily occurs at longer time scales, but there are also short-term effects of global warming on sea surface temperatures and temperature gradients. In addition, Wavelet coherence analysis also displayed the positive correlation between the global warming signal and the ENSO signal.

In summary, this paper investigates the effect of global warming on ENSO characteristics from the perspective of objective signals. There is an obvious positive correlation between global warming signal and ENSO signal, and global warming will promote the occurrence of ENSO phenomenon, especially the extreme La Niña events and extreme El Niño events. In the context of global warming, the frequency, intensity, and lifespan of ENSO occurrence will become larger and more inclined to more destructive El Niño events. As a result, it is urgent to strengthen international cooperation, conduct indepth mechanism research, gradually clarify the evolution and control mechanism of ENSO in the context of climate change, and curb the evolution process of global warming on ENSO.

CRediT authorship contribution statement

Zhiping Chen: Methodology, Software, Conceptualization. **Li Li:** Writing – original draft, Writing – review & editing, Validation. **Bingkun Wang:** Data curation. **Jiao Fan:** Formal analysis. **Tieding Lu:** Project Administration. **Kaiyun Lv:** Funding Acquisition.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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Appendix A. Basic principles of independent component analysis

The basic model of ICA can be described by Eq. (1), which illustrates how the observed signal is formed from the source signal (S) with the influence of noise (Cardoso J F, 1998). The intention of ICA is data mining which means that the mixing matrix (A) and the source signal (S) can be estimated simultaneously in case of the observed signal (X) is known.

$$X = AS \tag{1}$$

where A is the mixing matrix of dimension $m\times\ n(m\leq n)$. The matrix form of observed signal (X) can be described by $X = (x_1(t), x_2(t), \cdots, x_i(t), \cdots, x_m(t))^T$, m represents the sum number of observed signals. Similarly, the matrix form of source signal (S) can be described by $S = (s_1(t), s_2(t), \cdots, s_i(t), \cdots, s_n(t))^T$, n stands for the sum number of source signal (Hyvärinen and Oja, 2000). Moreover, the vectors of $s_i(t)$ and $s_{i+1}(t)$ are statistically independent of each other. FastICA approach (Bingham and Hyvärinen, 2000) is employed to separate the independent components of specific humidity monthly anomalies signals.

A.1. Basic principles of the non-parametric method

The optimal dimension selection formula based on the non-parametric method is as follows (Koch and Naito, 2007):

$$\widehat{p}_k = argmax_{2\leq p\leq p^*}\widehat{I}_k(p), \quad k = 3, 4 \tag{2}$$

In the formula, $p^* = min\{\mathrm{N}, \mathrm{d}\}$where Nand d are the sample number and dimension of matrix data respectively; $k = 3, 4$represents the optimal dimension selection criteria based on skewness and kurtosis respectively (Koch and Naito, 2007), where the formula of skewness or kurtosis is shown in Eq. (3); The optimal output dimension $\widehat{p}_k$ is determined by the maximum deviation value $\widehat{I}_k(p)$ across all dimensions.

The skewness ($k = 3$) and kurtosis ($k = 4$) can be calculated by the following formula:

$$\widehat{B}_k(\alpha|X_1, \ldots, X_N) = \left|\frac{1}{N}\sum_{i=1}^{N}\left\{\frac{\alpha^T(X_i-\overline{X})}{\sqrt{\alpha^T S\alpha}}\right\}^k - 3\delta_{k,4}\right| \tag{3}$$

where $\alpha \in \mathscr{S}^{d-1}$, $\mathscr{S}^{d-1}$ is the unit sphere in $\mathbb{R}^d$; X_i is the random variable; $\overline{X}$ denotes the sample mean and S is the sample covariance matrix; and $\delta_{k,4}$ is the Kronecker delta.

N is the sample number of Matrix data; $\widehat{\beta}_k(Z(p)_1, \dots, Z(p)_N)$ represents the maximum skewness ($k = 3$) or kurtosis ($k = 4$) of N random variables $Z(p)_i$ under dimension p; $\widehat{UB}_k(p)$ is the upper bound of $\sqrt{\frac{N}{k!}}\widehat{\beta}_k(Z(p)_1, \dots, Z(p)_N)$.

The deviation value is calculated by the following formula:

$$\widehat{I}_k(p) = \sqrt{\frac{N}{k!}}\widehat{\beta}_k(Z(p)_1, \dots, Z(p)_N) - \widehat{UB}_k(p) \tag{4}$$

where N is the sample number of Matrix data; $\widehat{\beta}_k(Z(p)_1, \dots, Z(p)_N)$ represents the maximum skewness ($k = 3$) or kurtosis ($k = 4$) of N random variables $Z(p)_i$ under dimension p; $\widehat{UB}_k(p)$ is the upper bound of $\sqrt{\frac{N}{k!}}\widehat{\beta}_k(Z(p)_1, \dots, Z(p)_N)$.

The upper bound in Eq. (5) is calculated by the following formula:

$$\begin{aligned}\widehat{UB}_k(p) = &\sum_{e=0}^{p-1} \omega_{p-e,k}\widetilde{\Lambda}_{p-e,\rho-p+e}(tan^2\theta_k|T_e) \\ &+\left(\frac{2k-2}{3k-2}\right)^{1/2} E[\chi_\rho][1-\Psi(\theta_k,p)]\end{aligned} \tag{5}$$

where e is an even number from 0 to $p-1$; χ_ρ represents the random variable of χ distribution with degree of freedom ρ; $E[\chi_\rho]$ represents the mathematical expectation of the random variable. In the formula, $\theta_k = cos^{-1}\left[\left(\frac{2k-2}{3k-2}\right)^{1/2}\right]$; T_e is the upper quantile of χ^2 distribution; $\omega_{p-e,k}$, $\widetilde{\Lambda}_{p-e,\rho-p+e}(tan^2\theta_k|T_e)$ and $\Psi(\theta_k,p)$ are expressed by Eq. (6–8, and) respectively.

$$\omega_{p-e,k} = (-1)^{e/2}k^{(p-1)/2}\left(\frac{k-1}{k}\right)^{e/2}\frac{\Gamma[(p+1)/2]}{\Gamma[(p+1-e)/2](e/2)!} \tag{6}$$

where $\Gamma[X]$ is the cumulative probability distribution function value of gamma distribution of X.

$$\widetilde{\Lambda}_{m,n}(b|T) = \frac{T}{N}\sum_{j=1}^{N} F_U\left(\sqrt{\gamma_j}\right) G_n(b\gamma_j) \tag{7}$$

In the formula, $m = p - e$, $n = \rho - p + e$, $b = tan^2\theta_k$, $T = T_e$; N is the sample number of Matrix data; $\gamma_1, \dots, \gamma_N$ is an independent identically distributed random variable from the ${\chi_m}^2$ distribution; F_U represents the cumulative distribution function of random variable U, which is uniformly distributed on the interval $[0, T]$; G_n represents the cumulative distribution function of ${\chi_n}^2$.

$$\Psi(\theta_k,p) = \sum_{e=0}^{p-1}\omega_{p-e,k}\overline{\mathscr{B}}_{(p-e)/2,(\rho-p+e)/2}\left(\frac{2k-2}{3k-2}\right) \tag{8}$$

where $\overline{\mathscr{B}}_{\alpha,\beta}$ represents the upper quantile probability of beta distribution; $\alpha = (p-e)/2$, $\beta = (\rho - p + e)/2$.

Appendix B. Signal extraction based on ERA5 specific humidity and ICA

According to the preceding analysis, to obtain high-quality signals of ENSO and global warming, it is imperative to consider the impact of nonlinearity and output dimensionality on signal quality. In order to overcome these challenges, this study proposes the use of the strategy of ICA combined with a non-parametric method for signal extraction. The detailed signal extraction procedure is illustrated in Fig. B1.

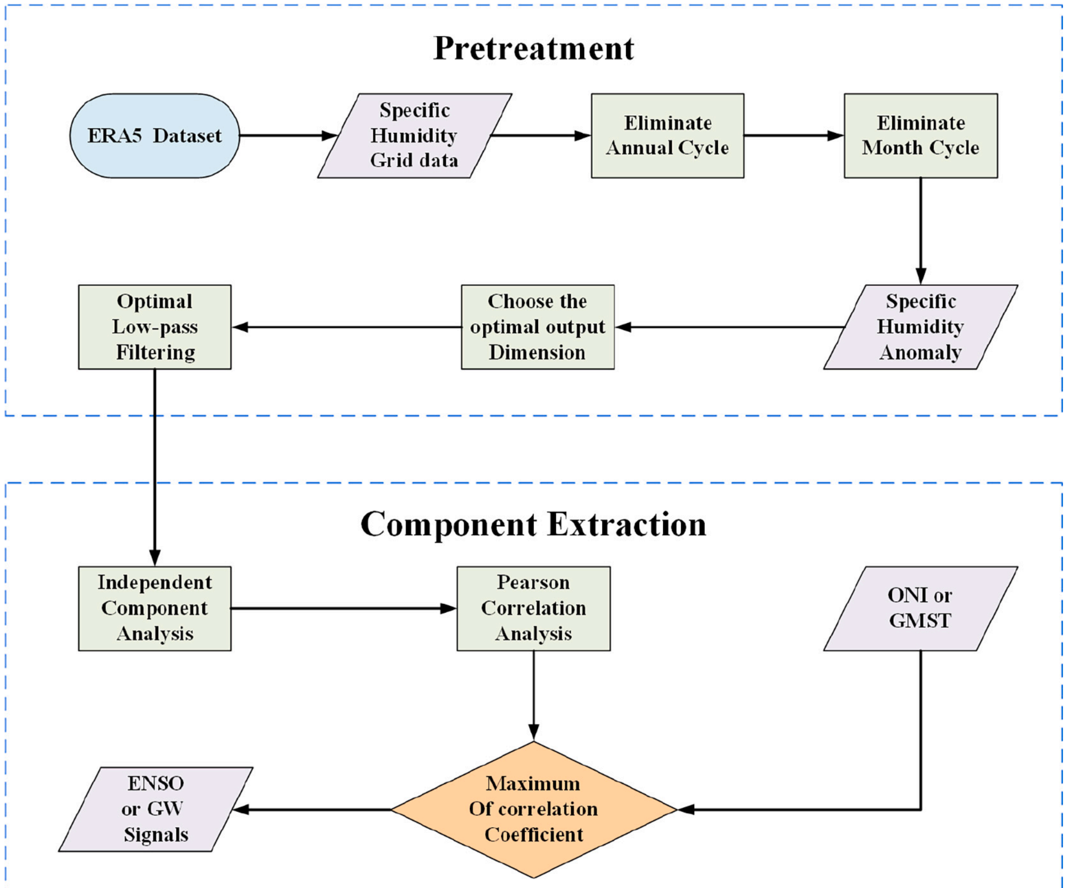

Fig. B1. Extraction processing flow of ENSO and Global Warming Signals.

The process of extracting ENSO and global warming signals can be divided into two parts: pretreatment and component extraction. The pretreatment primarily consists of the following four steps. Firstly, the grid specific humidity data is obtained from the ERA5 dataset. Secondly, the specific humidity monthly anomalies signals were extracted by eliminating the annual and month-to-month variations in the daily specific humidity time series (Sun and Lan, 2021). In detail, the annual mean anomalies were obtained by subtracting the month-by-month climatic mean state, on the other hand, the 1–2-1 three-point binomial filter was used to eliminate the monthly variability (Wang et al., 2012). Thirdly, the optimal output dimension is determined by a nonparametric method based on the specific humidity monthly anomaly signal (Koch and Naito, 2007), and the optimal output dimension obtained from the specific humidity monthly anomaly data in this paper is 2. Fourthly, in order to eliminate the noise of specific humidity monthly anomalies signals, the optimal low-pass filtering was applied for those signals (Chen et al., 2018). However, due to the uncertainty of the optimal cut-off frequency of low-pass filtering, it is essential to compare the results with different cut-off frequencies to determine the optimal cut-off frequency of low-pass filtering.

According to the selected optimal output dimension, independent components (ICs) can be estimated by ICA for all filtering cut-off frequencies. For each filtering cut-off frequency, the Pearson correlation analysis was carried out between the ICs and ONI/GMST. The optimal cut-off frequency of the low-pass filter was determined when the maximum absolute value of the correlation coefficient was obtained. The time series of ICs which have the maximum absolute value of the correlation coefficient with ONI were regarded as ENSO signals. Similarly, the time series of ICs which have the maximum absolute value of the correlation coefficient with GMST were regarded as global warming signals.

Considering the auto-correlation between the time series of extracted ICs and ONI/GMST, the Pearson correlation analysis should be carried out under the student's *t*-tests at the 95% confidence level (Oort and Yienger, 1996). And the actual degrees of freedom of the sample can be described by Eq. (9) at the moment.

$$n_{df} = n\left[1 + \frac{2}{n}\sum_{\tau=1}^{n-1}(n-\tau)r_\tau(x)r_\tau(y)\right]^{-1} \tag{9}$$

where n_{df} is the actual degree of freedom of the sample; $n$is the number of samples; $r_\tau(x)$and $r_\tau(y)$ are the autocorrelation coefficients of time series of ENSO/global warming signals and ONI/GMST after the lag time τ.

Then, the t-test quantity can be expressed by Eq. (10). The correlation coefficients in this paper meets the requirements of Student's t-tests at the 95% confidence level (Wallace et al., 1998).

$$t = \sqrt{n_{df} - 2}\,\frac{r}{\sqrt{1 - r^2}} \tag{10}$$

where t is the t-test quantity; n_{df} is the actual degree of freedom of the sample; r is the Pearson cross correlation coefficient.

B.1. The signal extraction results

To provide a clearer understanding of the signal extraction process, the results of signal changes after each step are presented here.

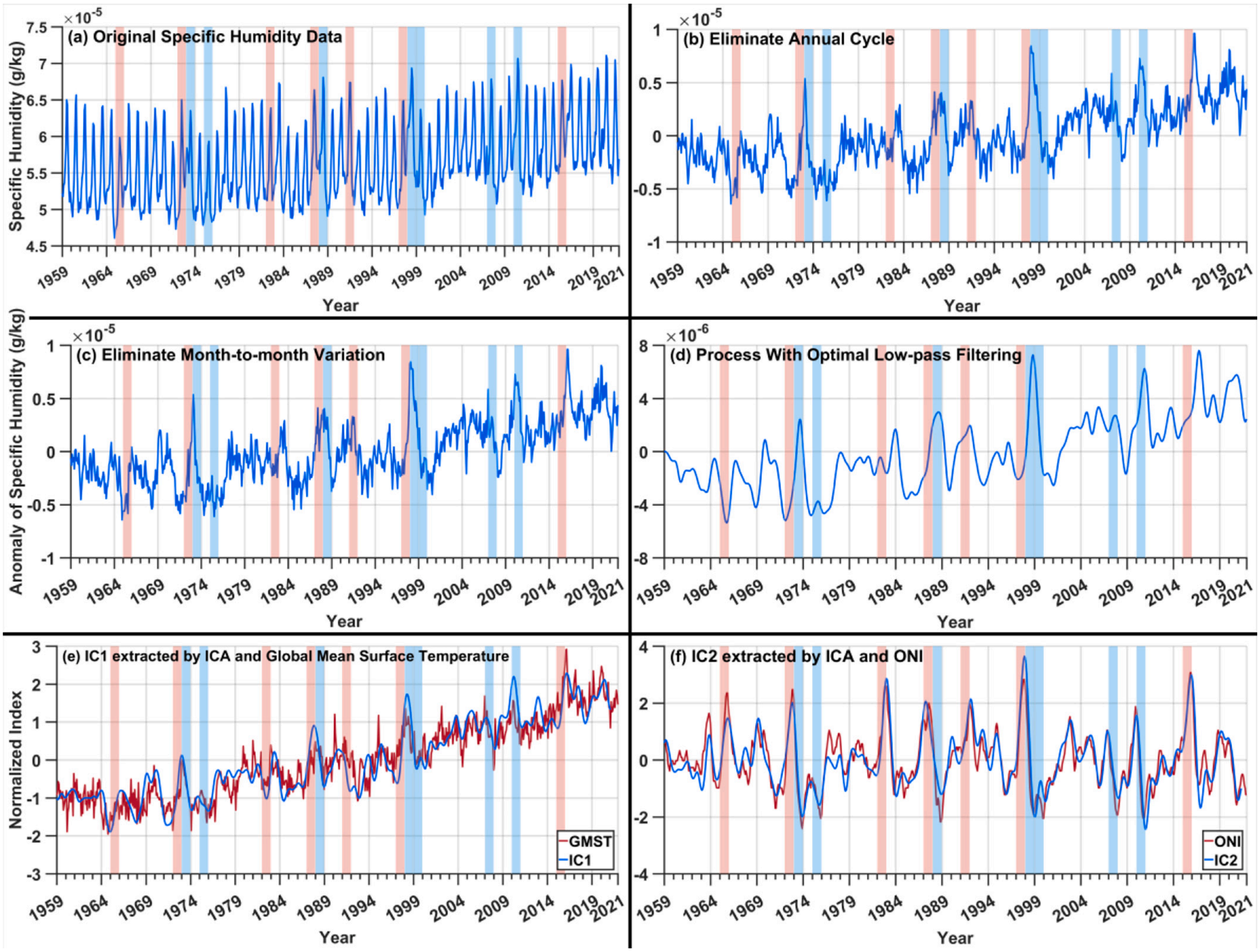

Fig. B2. Global warming and ENSO signal extraction process. (a) Global mean specific humidity; (b) Deducting the annual cyclic variation of the global mean specific humidity; (c) Deducting the inter-month variation based on (b); (d) Processing of time series with optimal low-pass filter based on (c); (e) ENSO signal extracted by the strategy of ICA combined with a non-parametric method; (f) Global warming signal extracted by the strategy of ICA combined with a non-parametric method.

The results of the signal extraction process are shown in Fig. B2. Fig. B2a displays the original specific humidity data, from which it can be seen that there is a significant annual cycle signal in the global average specific humidity data. To overcome the influence of annual cycle signal, Fig. B2b presents the results after subtracting the month-by-month climatic mean state to remove the annual cycle. Similarly, to mitigate the impact of monthly variability, the 1–2-1 three-point binomial filter was used to eliminate the monthly variability (Fig. B2c). Additionally, to further remove the influence of unknown factors, the optimal low-pass filter was applied (Fig. B2d). It can be observed that the optimal low-pass filtering effectively eliminates the noise. Finally, the results of the global warming and ENSO signals extracted by the strategy of ICA combined with a non-parametric method in Fig. B2e and Fig. B2f, respectively.

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